

YUBO ZHAO

Software Engineering | Machine Learning | Quantitative Research | Robotics
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EDUCATION

University of Waterloo - Bachelor of Software Engineering 1A Software Engineering Waterloo, Ontario	Expected 2031
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EXPERIENCE & COMPETITIONS

Team Canada Member - International Olympiad in Artificial Intelligence (IOAI) <i>Competitive Artificial Intelligence</i> - Selected to represent Team Canada at IOAI 2025; prepared across machine learning, deep learning, computer vision, NLP, data preparation, and model optimization. - Applied algorithmic reasoning, experimental design, and performance evaluation to advanced AI problem settings under competitive constraints.	2025
Robotics Team Member - FIRST Tech Challenge World Championship <i>FTC World Championship - 2nd Place Globally</i> - Collaborated across software, mechanical, electrical, and strategy functions to develop and optimize a competition robot for autonomous and driver-controlled performance. - Applied iterative engineering, sensor integration, autonomous-control logic, debugging, and performance tuning in a high-pressure international competition environment.	2024
Machine Learning / Quantitative Research Intern <i>Shaanxi Shun Yicheng Investment Management Co., Ltd. Xi'an, China</i> - Supported financial machine-learning workflows including preprocessing, feature engineering, model evaluation, signal validation, and performance benchmarking. - Investigated predictive modeling, risk analysis, and backtesting with emphasis on leakage prevention, generalization, and overfitting control.	Jul-Aug 2025

SELECTED PROJECTS

Brain Tumor Detection System | *Python | PyTorch | MiniMaxViT | Medical Imaging*

- Developed an MRI classification pipeline using a compact vision-transformer architecture; tested preprocessing, augmentation, oversampling, class balancing, and validation strategies.
- Evaluated precision, recall, F1 score, confusion matrices, generalization, and potential dataset artifacts rather than relying on accuracy alone.

Gesture-Controlled Drawing Interface | *Python | OpenCV | MediaPipe | scikit-learn*

- Trained a Random Forest classifier on 21-point hand-landmark features to recognize open, closed, and pointing gestures with 98% test accuracy.
- Implemented real-time drawing, erasing, temporal smoothing, gesture-state logic, latency controls, tool switching, snapshot capture, and precise pointing coordinates.

Jarvis - Local Multimodal AI Assistant | *Python | Ollama | OpenCV | Speech | APIs*

- Built a modular local assistant combining LLM inference, speech-to-text, text-to-speech, memory, external APIs, and real-time computer vision across voice, text, and camera input.

Future-Sim - World Model Prototype | *Python | NumPy | Pygame*

- Created a configurable simulation framework for world-state representation, environment dynamics, future-state generation, and outcome estimation; explored predictive world-model reasoning.

TECHNICAL SKILLS

Programming & Software: Python, C++, Java, JavaScript, SQL, HTML/CSS, Git/GitHub, Linux, VS Code, REST APIs, OOP, data structures and algorithms

Machine Learning & Data: PyTorch, scikit-learn, NumPy, pandas, Matplotlib; CNNs, ResNet, Vision Transformers, YOLO, transfer learning, fine-tuning, cross-validation, feature engineering, precision/recall/F1, ROC-AUC

AI / CV / Robotics: OpenCV, MediaPipe, local LLM/Ollama integration, RAG fundamentals, multimodal systems, speech recognition, sensor integration, PID control, IMU and sensor-fusion fundamentals

HONOURS

Euclid Mathematics Contest 2026 - Honour Roll, 86/100 | CSMC 2025 - Honour Roll, 48/60 | Sir Isaac Newton Physics Contest - Top 5% Globally